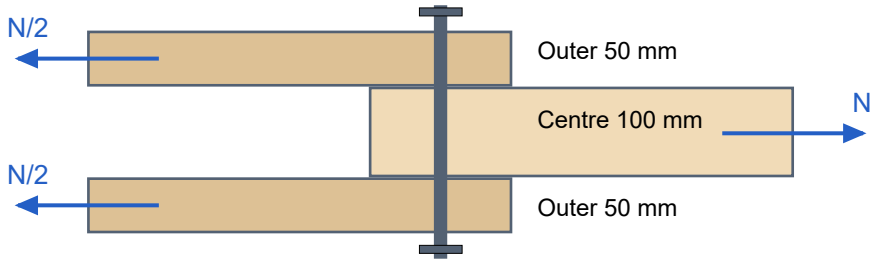
A symmetric bolted timber joint

18 October 2026 | Version 4 | AI-assisted educational study

A reproducible ambient-temperature calculation study linking force transfer, Johansen resistance, detailing, movement and a proposed laboratory investigation.

Scope: educational component results only. No overall connection approval, construction design, physical testing, calibrated finite-element model or completed personal study is claimed.



One M12 bolt; two shear planes; width 200 mm

200 mm overlap; bolt 100 mm from each loaded free end

Conceptual baseline detail, not to scale or for fabrication

Definition and prescribed example

The central C24 member transfers the full tensile action into two identical outer C24 members. Each outer member carries half the action. Balanced load introduction and identical outer members are required assumptions. There is one bolt, one transfer station and no fastener-group reduction.

Quantity	Selected input
Timber geometry	Central thickness 100 mm; each outer thickness 50 mm; width and overlap 200 mm
Bolt and washers	M12 grade 8.8; Ø13 mm hole; two Ø40 × 4 mm washers
Material references	C24: characteristic density 350 kg/m³; mean density 420 kg/m³; parallel tensile strength 14.5 MPa
Environment and model	Service class 2; load parallel to grain; smooth bolt shank at both planes; zero friction, pretension and rope effect

C24 values are reference properties, not product declarations. Bolt tensile strength 800 MPa is a specified study assumption. Sources: [1], Table 3.3, p.10 and pp.44-45. Full references appear on page 8.

Actions and Johansen resistance

Permanent tensile action is 3 kN and variable tensile action is 2 kN. The action factors are illustrative study inputs, not a verified project combination. Compression reversal, bending and out-of-plane force are excluded.

$$N_{Ed} = 1.35(3) + 1.5(2) = 7.05 \text{ kN}, \qquad N_{ser} = 3 + 2 = 5.00 \text{ kN}$$

$$f_{h,0,k} = 0.082(1 - 0.01d)\rho_k = 25.256 \text{ MPa}$$

$$M_{y,Rk} = 0.3 f_{u,k} d^{2.6} = 153.491 \text{ kN mm}$$

The outer thickness is t_1 = 50 mm and central thickness t_2 = 100 mm. Identical materials give an embedment-strength ratio β = 1. The following values are per bolt and **per shear plane**.

$$F_{g,Rk} = f_{h,1,k} t_1 d = 15.154 \text{ kN}$$

$$F_{h,Rk} = 0.5 f_{h,2,k} t_2 d = 15.154 \text{ kN}$$

$$F_{j,Rk} = \frac{1.05 f_{h,1,k} t_1 d}{2+\beta}\left[\sqrt{2\beta(1+\beta) + \frac{4\beta(2+\beta)M_{y,Rk}}{f_{h,1,k} d t_1^2}} - \beta\right] = 8.146 \text{ kN}$$

$$F_{k,Rk} = 1.15\sqrt{\frac{2\beta}{1+\beta}}\sqrt{2M_{y,Rk} f_{h,1,k} d} = 11.092 \text{ kN}$$

Mode j governs the selected model. Apply the two-plane factor once, with k_{mod} = 0.8 and γ_M = 1.3:

$$R_d = 2\frac{k_{mod}}{\gamma_M}\min(F_{g,Rk}, F_{h,Rk}, F_{j,Rk}, F_{k,Rk}) = 10.026 \text{ kN}$$

Component utilisation: 0.703 for the 7.05 kN action and 10.026 kN resistance. This is not an overall connection pass. Brittle timber failure, bolt steel resistance, washer bearing and installation remain separate checks.

Sources: [1], pp.36-38 and 44; [3], EN 1995-1-1 8.2.2, Eq.8.7; 8.5.1.1, Eqs.8.30-8.33.

Detailing and nominal net tension

Checks use the baseline parallel-to-grain geometry. Both loaded free ends are assessed because the central and outer members terminate on opposite sides of the bolt.

Check	Actual / mm	Requirement / mm	Result
First loaded end	100.0	≥ 84.0	PASS
Opposite loaded end	100.0	≥ 84.0	PASS
Both centred edges	100.0	≥ 36.0	PASS
Timber hole diameter	13.0	≤ 13.0	PASS
Washer outer diameter	40.0	≥ 36.0	PASS
Washer thickness	4.0	≥ 3.6	PASS
Washer radius / available face	20.0	≤ 100.0	PASS
Washer outer diameter / timber hole	40.0	> 13.0	PASS

Inter-fastener and inter-row spacing do not apply to one bolt. Minimum distances and footprint geometry do not establish timber splitting, end tear-out or washer bearing resistance. Washer-hole dimensions and full product geometry remain unverified.

Nominal net-section tensile check

$$A_{net} = (b - d_0)t, \qquad R_{t,d} = A_{net}\frac{k_{mod} f_{t,0,k}}{\gamma_M}$$

No favourable tensile size increase is used. The central member takes the full action; each outer member takes half. This nominal-section check does not model fracture, splitting or shear-out.

Member	Net area / mm²	Demand / kN	Resistance / kN
Centre	18,700	7.050	166.862
One outer member	9,350	3.525	83.431

Both nominal net-section utilisations are 0.0423. Sources: [1], pp.22 and 45; [3], 6.1.2, Table 8.4 and 10.4.3.

Joint movement and duration envelope

The two identical shear-plane springs act in parallel. Stiffness uses mean density 420 kg/m³ and describes a post-seating linear approximation, not the total measured response of a fabricated joint.

$$K_{ser,\,plane} = \frac{\rho_m^{1.5} d}{23} = 4.491 \text{ kN/mm}, \qquad K_{joint} = 2K_{ser,\,plane}$$

$$u_{inst} = \frac{N_{ser}}{K_{joint}} = \frac{5.00}{8.9817} = 0.5567 \text{ mm}$$

Sustained-load movement

For identical service-class-2 timbers, the selected first-generation model uses member k_{def} = 0.8 and connection factor $2k_{def}$ = 1.6. The quasi-permanent variable-action factor ψ_2 = 0.3 is prescribed for this study and requires project confirmation.

$$u_{fin,\,elastic} = \frac{G_k(1 + 2k_{def}) + Q_k(1 + 2\psi_2 k_{def})}{K_{joint}} = 1.1980 \text{ mm}$$

Contribution	Movement / mm
Permanent action, including its creep increment	0.8684
Variable action, including its reduced creep increment	0.3296
Final elastic total	1.1980
Prescribed additional free play in baseline	0.0000

The instantaneous characteristic variable action remains in the expression; only its creep increment is reduced by ψ_2. Free play is an explicit total allowance, not inferred from the hole diameter; it is added once without creep amplification. The 2 mm movement budget is an illustrative choice, not a code limit.

Separate load-duration cases

Case	Design action / kN	k_{mod}	Resistance / kN	Utilisation
Combined G + Q	7.050	0.8	10.026	0.703
Permanent G only	4.050	0.6	7.520	0.539

The combined case governs this selected envelope. It is not a complete set of project combinations. Sources: [1], pp.7-8 and 32-33; [2], pp.157-159, Table 5.4/Eq.5.86; [3], 2.2.3(5), 2.3.2.2(3) and 7.1/Table 7.1.

Controlled parameter sensitivity

Analytical comparisons hold the model topology and material fixed. The diameter series sets hole diameter to d + 1 mm while retaining the baseline end distances and washers. These are controlled comparisons, not optimisation or measured behaviour.

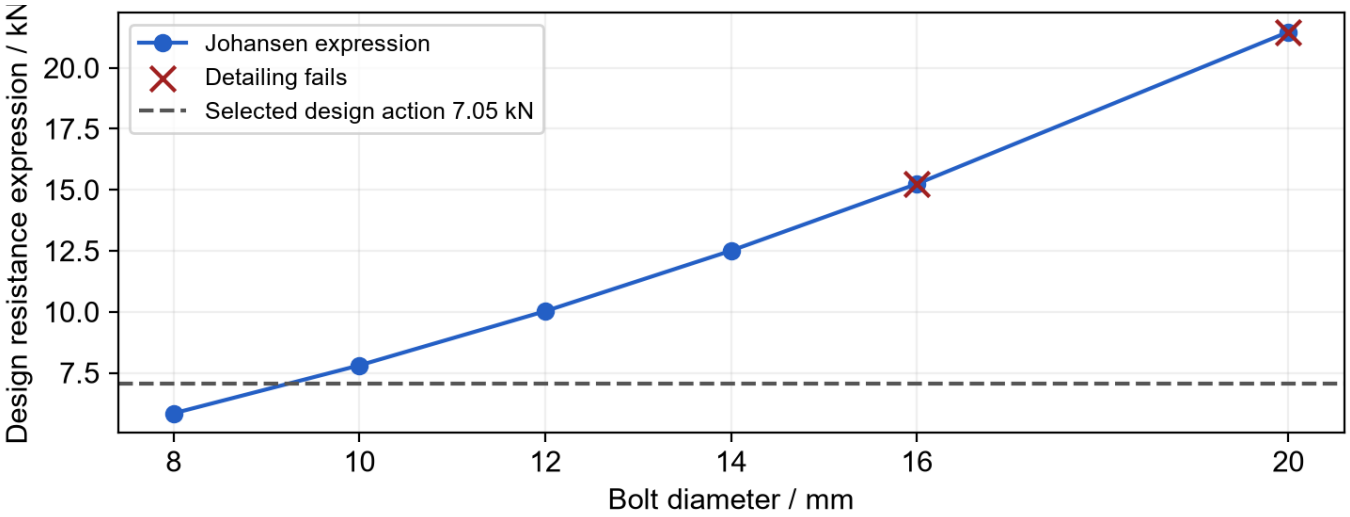


Figure 2. A larger Johansen expression does not imply acceptable detailing. Crosses identify scenarios with failed geometry checks.

Diameter / mm	Expression / kN	Utilisation	Geometry	Final slip / mm
8	5.840	1.207	PASS	1.797
10	7.807	0.903	PASS	1.438
12	10.026	0.703	PASS	1.198
14	12.505	0.564	PASS	1.027
16	15.242	0.463	FAIL	0.898
20	21.472	0.328	FAIL	0.719

For M8, the selected action exceeds the resistance. M16 and M20 fail the fixed detailing. The baseline M12 model has no failure in the checked components; unresolved mechanisms still prevent an overall pass.

Free play and sustained action

For M12, adding 0.5, 1.0 and 2.0 mm total free play changes final movement from 1.198 mm to 1.698, 2.198 and 3.198 mm. Thus 1.0 mm free play exceeds the chosen 2 mm budget. Varying ψ_2 from 0 to 1 gives final elastic movements of 1.091 to 1.447 mm. These are prescribed scenarios, not measured tolerances.

Source: generated full-precision sensitivity JSON/CSV accompanying this release; equations and boundaries are those on pages 2-4.

A proposed laboratory investigation

Status: proposed only. No specimens have been fabricated or tested. The objective is to compare the predicted mechanism and post-seating stiffness with observations and identify where the idealisation fails.

Series	Single change from baseline	Repeats	Question
A	M12; 50 mm outer members	3	Does the symmetric model reproduce local slip and observed failure?
B	M10; Ø11 mm hole	3	What changes at fixed member dimensions?
C	35 mm outer members; M12	3	Does outer thickness change the limiting mechanism?

Nine tests form a proposed exploratory programme. This sample size does not establish characteristic resistance or statistical population coverage. Record density, moisture, defects and actual geometry for every member.

Rig and instrumentation

1. Use a balanced clevis or equalising fixture to load the centre against both outer members. Laboratory staff must review grips, restraint, capacity and guarding.

2. Measure force with a calibrated load cell. Use paired local displacement transducers on both sides to distinguish relative slip, rotation and outer-member imbalance.

3. Record actuator displacement separately because it includes fixture and member deformation. Photograph seating, ovalisation, cracks, embedment and post-test bolt deformation.

Procedure and retained evidence

1. Measure as-built geometry and material properties; check alignment and instrument range. Run a pilot to establish an appropriate capacity estimate.

2. Finalise loading history, rate and analysis windows against the licensed EN 26891 text and approved laboratory procedure before testing.

3. Separate seating from post-seating response. Report declared stiffness windows, peak force and observed failure mode. Compare each specimen with calculations using its own recorded properties. Retain raw data, scripts, photographs and anomalous tests.

EN 26891 is the relevant strength/deformation framework [5]. EN 12512 concerns a separate later cyclic programme [6]. Only their public scope pages were checked; complete procedures have not been implemented.

Assembly, wider performance and learning

Manufacturing and disassembly evidence to collect

1. Prepare coordinated drawings identifying grain, loaded free ends, dimensions, hole alignment, shank position and accessible head/nut locations. Confirm grading and bolt/washer certificates.

2. Fabricate a trial detail under workshop supervision. Inspect drilling damage, fit, washer seating and balanced assembly. Confirm tightening without unverified friction credit.

3. Record assembly time, tools, access and defects. After disassembly, inspect holes, cracks and bolt straightness. Any reuse proposal needs measured condition and appropriate testing.

A proposed coupon inventory uses one 100 × 200 × 600 mm central piece and two 50 × 200 × 600 mm outer pieces: 0.024 m³ nominal timber before waste. The 35 mm outer variant uses 0.0204 m³. The 600 mm lengths are planning assumptions, not prescribed standard dimensions. Grip and boundary-condition design remain open.

Sustainability, envelope and fire boundaries

An environmental comparison requires product-specific environmental declarations, actual mass, transport, machining waste, fixings and an agreed life-cycle boundary. No life-cycle assessment or quantified environmental saving is claimed. Reuse and demountability benefits need evidence.

A renovation-panel detail must also coordinate insulation, sheathing, moisture control and the existing-building load path. Fire exposure, protection continuity and connection performance need a separate assembly assessment. The toolkit’s isolated-member section-loss illustration cannot provide this.

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Files were prepared with AI assistance. Personal review and reproduction have not been recorded. Complete and date these exercises before presenting them as personal understanding:

1. Recalculate all four modes and explain why the two-plane factor is applied once.

2. Sketch the full and half-force load paths; distinguish one transfer station from two in series.

3. Explain instantaneous deformation, creep, free play and why a larger bolt may fail detailing.

4. Identify which measurements could challenge symmetry, stiffness and governing-mode assumptions.

A later Abaqus study requires explicit timber constitutive modelling, contact, bolt representation, mesh assessment and a calibration/validation split. No Abaqus input or solver job is part of this case.

Verification, sources and limits

Implementation checks

The release records 165 passing automated tests. New Python/browser numerical parity covers 40 vectors, 3,696 numeric comparisons, 1,381 classification comparisons and 12 expected rejections, with no failures. These checks establish implementation agreement within their fixtures; they do not establish physical validity. Browser interface behaviour was not verified for this release.

The source and full-precision input, result, sensitivity and parity files accompany the local v4 release. This PDF is a static reading copy. The existing live toolkit is at gitmeysambayat.github.io/TimberFacadeConnectionToolkit.

Unresolved engineering checks

Current standard edition, National Annex and project actions; product declarations; bolt steel resistance and washer bearing; splitting, shear-out and block failure; rotational restraint; fabrication, seating and tightening; moisture, corrosion and durability; cyclic degradation and repeated assembly; connection and assembly fire resistance; physical testing and long-term model validation all remain outside the completed scope.

Sources and selected locators

[1] Swedish Wood/TDUK. Design of timber structures, Volume 2, UK edition 1:2022. Properties p.10; factors pp.7-8; net tension p.22; slip pp.32-33; resistance pp.36-38; bolts/detailing pp.44-45. Source link.

[2] Swedish Wood. Design of timber structures, Volume 1, edition 3:2022. Connection creep pp.157-159, Table 5.4 and Eq.5.86. Source link.

[3] BS EN 1995-1-1:2004+A1:2008, publicly hosted copy. Selected first-generation clause cross-check, not the current official standards service. Source link.

[4] BSI. UK National Annex and Eurocode coexistence statement. The study has not checked every applicable National Annex choice and does not implement second-generation 2025 provisions. Source link.

[5] BSI. BS EN 26891, scope of strength/deformation testing for mechanically fastened timber joints. Source link.

[6] BSI. BS EN 12512, scope of cyclic testing for mechanically fastened timber joints. Source link.

Overall status: EDUCATIONAL COMPONENT RESULTS. NO OVERALL DESIGN PASS.